Student Handout

Name: _____ **Date:** _____ **Class:** _____

MyDesign Portfolio

Element D Initial Template AT - 15 points

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1

Date: 04/07/2025
Description: A program in MATLAB that lets you type in your classroom and shows the shortest path on a map from the Bulldog Lobby.
Drawing that includes distinguishing information such as scale, dimensions, and materials: Map with arrows from lobby to classroom.

Design Idea #2

Date: 04/07/2025
Description: A printed map with step-by-step written instructions next to it, like a guidebook.
Drawing that includes distinguishing information such as scale, dimensions, and materials: Map with numbered steps on paper.

Design Idea #3

Date: 04/07/2025
Description: A simple touchscreen map that shows directions when you press your room number.
Drawing that includes distinguishing information such as scale, dimensions, and materials: Touchscreen with buttons for each room.

Design Idea #4

Date: 04/07/2025
Description: A hallway poster with a QR code that opens a website showing directions.
Drawing that includes distinguishing information such as scale, dimensions, and materials: Poster with QR code and example phone screen.

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Design Idea Comparison and Selection *(D1, D2)*

(List your requirements from Element A below)	Weight (multiplied by each design score)	Design 1	Design 2	Design 3	Design 4
Accuracy	3	5	4	3	3
Speed	3	5	2	4	4
Map Clarity	2	5	3	4	2
Flexibility	2	4	2	3	3
Multi Floor	1	4	2	3	2
Total score	11	52	30	38	33

Our team chooses Design 1 **to solve the community problem because** it scored the highest in all of the design requirements.-

Design Blueprint *(D4)*

Design Blueprint #1

Date: 04/09/2025

Sketch or CAD Drawing (orthographic): Map with labeled hallways and stairways; nodes for rooms; arrows for paths.

Manufacturing Methods (cut, saw, paste, screw, tie, glue):

None (programming project)

Supply List and Costs:

None (programming project)

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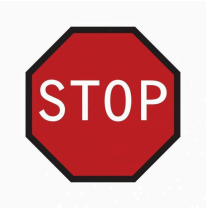
Plan of Action *(D3)*

- Week 1: Research map and room list
- Week 2: Import rooms into MATLAB
- Week 3: Write instructions for paths
- Week 4: Add algorithm for shortest path
- Week 5: Build user input
- Week 6: Test directions for rooms
- Week 7: Make adjustments based on feedback
- Week 8: Present final project



Iterations for the Solution (How did you change your prototype after building and testing it?):

Date of Iteration:

Description of Iteration:

Images if applicable:

Date of Iteration:

Name: _____ **Date:** _____ **Class:** _____

Description of Iteration:

Images if applicable:

Date of Iteration:

Description of Iteration:

Images if applicable:

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was *comprehensive, iterative, and consistently* defensible (viable design highly likely)	was *thorough, iterative, and* generally defensible (viable design likely)	was *adequate, generally* iterative and defensible *(viable design is possible)*	was *partial or overly general and only somewhat* iterative and/or defensible (raising issues with the design viability)	*was incomplete or minimally iterative and/or defensible (raising issues with the design viability)*	*was not evident that there was an attempt*

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2.Design Solution Choice. The Design Solution chosen _____.	was *well-justified* and demonstrated attention to *all* design requirements	*was justified* and demonstrated attention to *most if not all* design requirements	*was explained with reference to at least some* design requirements	*was not sufficiently explained with reference to* design requirements	*lacked support related to* design requirements	*was not attempted or evidence not provided*
3. Plan of Action. The plan of action _____.	*has considerable merit and would easily* support repetition and testing for effectiveness by others	*would* support repetition and testing for effectiveness by others	*might not clearly or fully support repetition and testing for effectiveness by others*	*has insufficient detail to allow for testing for replication of results*	*is lacking to the point that the design solution cannot be tested*	*cannot be executed or does not exist*